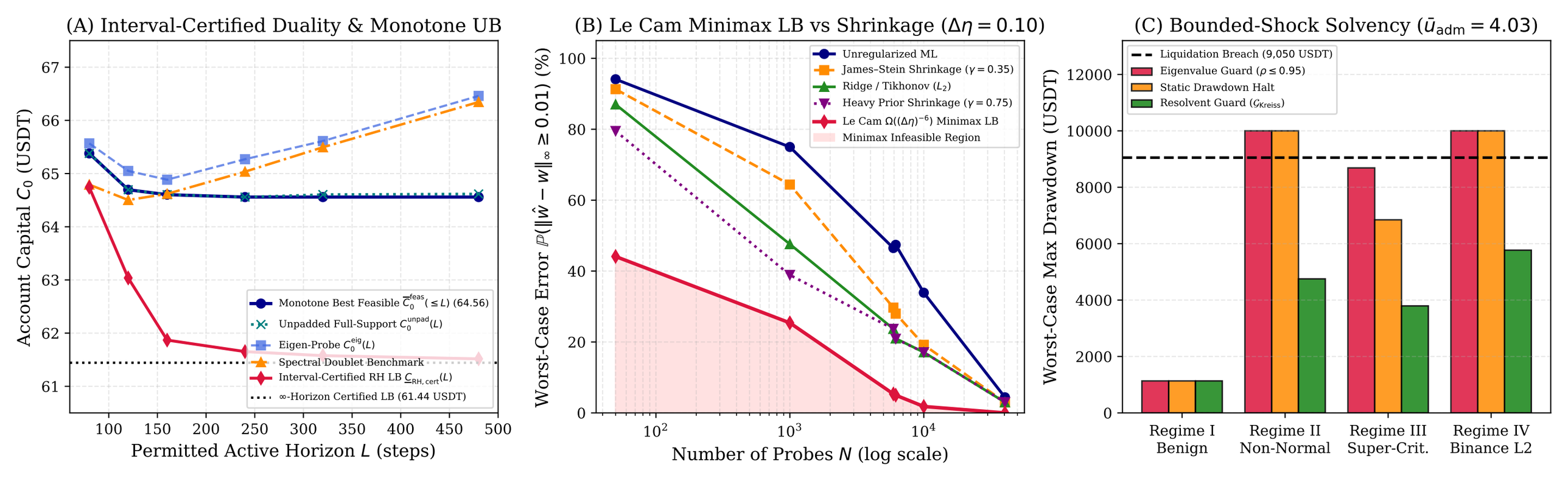

(A) Interval-Certified Duality & Monotone UB
Account Capital C_0 (USDT)
Permitted Active Horizon L (steps)
Monotone Best Feasible $\overline{C}_0^{\text{feas}}(\le L)$ (64.56)
Unpadded Full-Support $C_0^{\text{unpad}}(L)$
Eigen-Probe $C_0^{\text{eig}}(L)$
Spectral Doublet Benchmark
Interval-Certified RH LB $\underline{C}_{\text{RH, cert}}(L)$
∞-Horizon Certified LB (61.44 USDT)
(B) Le Cam Minimax LB vs Shrinkage ($\Delta\eta = 0.10$)
Worst-Case Error $\mathbb{P}(\|\hat{w} - w\|_\infty \ge 0.01)$ (%)
Number of Probes N (log scale)
Unregularized ML
James–Stein Shrinkage ($\gamma = 0.35$)
Ridge / Tikhonov (L_2)
Heavy Prior Shrinkage ($\gamma = 0.75$)
Le Cam $\Omega((\Delta\eta)^{-6})$ Minimax LB
Minimax Infeasible Region
(C) Bounded-Shock Solvency ($\bar{u}_{\text{adm}} = 4.03$)
Worst-Case Max Drawdown (USDT)
Liquidation Breach (9,050 USDT)
Eigenvalue Guard ($\rho \le 0.95$)
Static Drawdown Halt
Resolvent Guard ($\mathcal{G}_{\text{Kreiss}}$)
Regime I Benign
Regime II Non-Normal
Regime III Super-Crit.
Regime IV Binance L2